Subject Description Form

Subject Code	COMP 5122
Subject Title	E-Commerce Fundamentals and Development
Credit Value	3
Level	5
Pre-requisite/Exclusion	Nil
Objectives	The objectives of this subject are to: 1. introduce the infrastructure and functional components for e-commerce; 2. enable students to understand the enabling technologies for e-commerce; 3. study various e-commerce applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) acquire a thorough understanding of e-commerce and its applications, addressing complex interdisciplinary issues with professionalism; b) understand the enabling technologies for e-commerce, leveraging emerging technologies for global challenges; c) be able to generate critical review and consolidation of e-commerce, demonstrating critical awareness and integrity in making responsible decisions; and d) demonstrate leadership and qualities of reflective practitioners, fostering intercultural and interdisciplinary interactions.
Subject Synopsis/ Indicative Syllabus	• **Web system and programming**: Web system overview. HyperText Transfer Protocol (HTTP). Load balancing. Caching. HyperText Mark up Language (HTML). Client-side programming. Server-side programming. • **Cryptography:** Security requirements. Asymmetric key encryption. Symmetric key encryption. Message digest. Digital signature. Digital certificate. Public key infrastructure. • **Internet security:** IPSec. Firewall. Secure Socket Layer (SSL) Protocol/Transport Layer Security. Application layer security. • **Internet payment systems:** Secure electronic transaction (SET). Electronic cash. Electronic check. Micropayment methods. Smart card. • **E-commerce applications:** Business models. Consumer-oriented e-commerce. Business-oriented e-commerce. Auction. Case studies and examples. • **Advanced/current topics:** e.g., Mobile agent-based e-commerce, m-commerce.
Teaching/Learning Methodology	39 hours of class activities including lectures, tutorials, lab(s), workshop(s) and seminar(s) where applicable.

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<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific Assessment Methods/Tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td></tr>
<tr><td>Projects</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>Final Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr>
<tr><td>Total</td><td>100</td><td colspan="4"></td></tr>
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<tr><td>Student study effort expected</td><td>
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<tr><td colspan="2">Class Contact:</td></tr>
<tr><td>Class activities (lectures, tutorials, lab(s))</td><td>39 hours</td></tr>
<tr><td colspan="2">Other student study effort:</td></tr>
<tr><td>Self-study, project, exam</td><td>66 hours</td></tr>
<tr><td>Total student study effort</td><td>105 hours</td></tr>
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<tr><td>Reading list and references</td><td>
(1) Kenneth C. Laudon and Carol Guercio Traver, E-Commerce 2023-2024, Business, Technology and Society, 18th Edition, Pearson, 2023.
(2) Efraim Turban, Jon Outland, David King, Jae Kyu Lee, Ting-Peng Liang, and Deborrah C. Turban, Electronic Commerce: A Managerial and Social Networks Perspective, 9th Edition, Springer, 2022.
(3) Terry Felke-Morris, Web Development and Design Foundations with HTML5, 10th Edition, Pearson, 2023.
(4) Jon Duckett, JavaScript and jQuery: Interactive Front-End Web Development, 2nd Edition, Wiley, 2023.
(5) Frederik Voss, Blockchain and the Digital Economy: The Rise of Tokenomics, Springer, 2022.
(6) William Stallings, Cryptography and Network Security: Principles and Practice, 8th Edition, Pearson, 2023.
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